Surya Jothinathan

Udumalaipettai, Tiruppur, 642122

surya2003j@gmail.com (+91) 9600618481 linkedin.com/in/Surya

EXPERIENCE

Caliber Interconnect Solutions Pvt Ltd

PCB Designer Trainee *Nov 2025 - July 2026, Coimbatore*

- Designed and modified PCB layouts using Cadence Allegro PCB Designer for ATE Load Board applications.
- Updated schematics using OrCAD Capture CIS, ensuring design accuracy and compliance with project requirements.
- Performed component placement, routing, DRC verification, and PCB modifications to improve manufacturability and electrical performance.
- Generated manufacturing deliverables including Gerber files, drill files, fabrication drawings, assembly drawings, and BOM documentation.

ACADEMIC PROJECTS

IoT-Based Smart Water Consumption Monitoring and Billing System

- Designed an IoT-based water consumption monitoring system using ESP32 and a water flow sensor for real-time usage tracking and automated billing.
- Integrated sensing, power supply, and Wi-Fi communication hardware, and programmed ESP32 firmware to process flow data and transmit it wirelessly.
- Tools Used: ESP32, Water Flow Sensor, Arduino IDE, IoT, Wi-Fi Communication

Variable DC Bench Power Supply PCB Design

- Designed a variable DC bench power supply PCB with adjustable output voltage, incorporating voltage regulation, current limiting, and protection circuitry for testing and prototyping applications.
- Developed the complete schematic, PCB layout, power routing, and thermal management, applying DRC and DFM best practices along with manufacturing documentation.
- Tools Used: Cadence OrCAD Capture, Cadence Allegro

Automated Multi-Dose Medicine Dispensing System

- Designed and developed an automated medicine dispensing system to dispense multiple medications at scheduled times using a motorized dispensing mechanism and RTC
- Implemented sensor-based dispensing verification, missed-dose detection, and user alerts to improve medication adherence.
- Tools Used : ESP32, DS3231 RTC, Stepper Motor..

CERTIFICATIONS

- Altium Designer Essentials - Altium Education.
- Introduction to C - Infosys Springboard
- One-week Value-Added course on Programming in Arduino.
- PCB Designer Essentials - Great Learning

SUMMARY

Aspiring Hardware/PCB Design Engineer with practical experience in schematic design, PCB layout, design modifications, and manufacturing documentation. Looking to leverage my skills in innovative electronics product development while continuously learning advanced hardware design methodologies.

SKILLS

Languages	C
Tools	Cadence Allegro PCB Designer, OrCAD Capture CIS, KiCAD
PCB Design	Schematic Capture, PCB Layout Design, Component Placement, Routing, Multi-layer, DRC
Standards	DFM, DFA, IPC(Basics)
Soft Skills	Teamwork, Resilience.

EDUCATION

B.E - ECE

P.A College of Engineering and Technology

CGPA: 8.00/10.00

ACCOMPLISHMENTS

- Won 1st Prize in Project Presentation at P.A. College of Engineering and Technology
- Won 1st Prize in Paper Presentation at Hindustan Institute of Technology.
- Served as Secretary of the ISTE Student Chapter, coordinating technical events and student activities
- NSS Volunteer

LANGUAGES

- English (Fluent)
- Tamil (Native)